

---

# Nonlinear Tucker Decomposition with Polynomial Links

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

Tucker decomposition is a compact model for multiway data, but its map from a latent tensor to measurements is linear. Nonlinear measurement responses can therefore make a low-rank latent tensor appear higher-rank in the measured tensor. We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL), a Tucker enhancement that learns a polynomial link jointly with a Tucker backbone. NTD-PL keeps the backbone ranks fixed and adds only scalar link coefficients. Its polynomial powers induce a parameter-tied rank lift: each degree forms Veronese-lifted Tucker factors tied to the same latent tensor, rather than an untied high-rank correction.

We analyze this structure through Veronese rank expansion, separation from matched-rank Tucker, and approximation results for nonlinear responses. We optimize a masked reconstruction and completion objective by alternating Link Refresh, a small ridge update for the scalar link, with Linked-Backbone Descent and degree continuation. Experiments on controlled tensors, hyperspectral reconstruction and completion, rank-lift and Link Refresh diagnostics, and external cross-domain cases show scoped gains: NTD-PL helps when the residual left by matched-rank Tucker is coherent with a nonlinear response.

## 1 Introduction

Tensor decompositions provide compact models for multiway data in scientific data analysis, signal processing, and masked multiway learning [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013, Liu and Moitra, 2020]. Tucker decomposition is a standard choice because it separates a small core from mode-wise factors. It uses a linear map from the Tucker latent tensor to the measured tensor: the reconstructed Tucker value is fitted directly to the measurement. This assumption is natural for many data sets, but it can be mismatched to measurement or data-generation pipelines with response curves, saturation, calibration effects, or material-dependent nonlinearities [Debevec and Malik, 1997, Grossberg and Nayar, 2004, Bioucas-Dias et al., 2012, Heylen et al., 2014, Luo et al., 2024]. In such settings, the latent signal may be well organized by a low-rank Tucker tensor while the measured tensor appears to have larger multilinear rank after a nonlinear response.

A direct remedy is to increase Tucker rank, but this changes the multilinear budget and expands the core and factors throughout the model. More flexible nonlinear tensor decoders based on kernels, Gaussian processes, or neural networks can also be useful, but they address a broader problem and often make the relation to a Tucker backbone less explicit [Xu et al., 2011, Zhe et al., 2016, Liu et al., 2017, Saragadam et al., 2024, Ahn et al., 2024]. We ask a narrower question: can a Tucker model keep its chosen backbone ranks fixed while learning a response that explains coherent nonlinear residuals left by matched-rank Tucker?

We instantiate this idea with a polynomial link. Let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$$

37 be a Tucker latent tensor. Instead of fitting the measured tensor directly by  $\mathcal{S}$ , we model

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p},$$

38 where  $\odot p$  denotes the Hadamard power. We call the model Nonlinear Tucker Decomposition  
 39 with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case, keeps the Tucker  
 40 backbone and multilinear ranks fixed, and adds only  $P + 1$  scalar link coefficients. The higher powers  
 41  $\mathcal{S}^{\odot p}$  induce high-order interactions among the same core and factors, so the added expressivity is  
 42 parameter-tied to one learned latent tensor.

43 This viewpoint is central to the paper. NTD-PL is best understood as a Tucker enhancement and  
 44 shared-backbone rank lift, not as a generic nonlinear decoder. A degree- $p$  power of the latent tensor  
 45 behaves like an ordinary Tucker tensor whose factors are polynomial feature lifts of the original  
 46 factors, but those lifted features remain tied to one learned backbone. The new degrees therefore  
 47 reshape the low-rank geometry through the polynomial link, rather than adding an untied high-rank  
 48 tensor block. This is also why the link is not a post-hoc calibration layer.

49 Our claims are intentionally scoped. Increasing Tucker rank remains the right comparison when the  
 50 residual is ordinary multilinear structure, and we do not claim to dominate neural, kernel, or other  
 51 nonlinear tensor factorization methods. The intended regime is more specific: the residual left by  
 52 matched-rank Tucker should be coherent with a nonlinear response.

53 This scoped regime is also supported empirically. In controlled tensors, the gain grows with the  
 54 injected nonlinear residual energy and with the degree needed to approximate the response. On  
 55 real hyperspectral tensors, NTD-PL improves matched-rank Tucker in both full reconstruction and  
 56 held-out completion. More importantly, a one-step Link Refresh diagnostic predicts when NTD-PL  
 57 will help: across HSI scenes and non-HSI tensor units, larger diagnostic reduction is associated with  
 58 larger NTD-PL gains, while low-yield units show little or no benefit. These results suggest that  
 59 nonlinear responses are identifiable structure in several real tensor families.

60 **Contributions.** We make the following contributions.

- 61 • **Model.** We propose NTD-PL, a Tucker enhancement with a polynomial link on a Tucker latent ten-  
 62 sor. It keeps the Tucker backbone and multilinear ranks fixed while adding only scalar coefficients  
 63 for a parameter-tied response.
- 64 • **Theory.** We characterize the induced model class through Veronese rank expansion, separation  
 65 from matched-rank Tucker, and approximation results for nonlinear responses.
- 66 • **Optimization.** We derive a masked objective for reconstruction and completion, and optimize it  
 67 with Link Refresh, Linked-Backbone Descent, and degree continuation.
- 68 • **Experiments.** We evaluate controlled nonlinear residuals, hyperspectral reconstruction and com-  
 69 pletion, rank-lift and Link Refresh diagnostics, and external cross-domain boundary cases.

## 70 2 Related Work

71 **Low-rank tensor models and Tucker backbones.** CP, Tucker, tensor train, and tensor ring  
 72 decompositions provide compact representations for multiway data through different rank notions  
 73 and factor topologies [Kolda and Bader, 2009, Sidiropoulos et al., 2017, De Lathauwer et al., 2000,  
 74 Oseledets, 2011, Zhao et al., 2016]. Recent completion work has improved scalability, sampling  
 75 guarantees, and visual-data priors while still mostly changing the tensor rank model, tensor-network  
 76 topology, or low-rank regularizer [Liu and Moitra, 2020, Wang et al., 2021, Zheng et al., 2021].  
 77 Tucker is the closest baseline for NTD-PL because it uses the same core-factor multilinear backbone.  
 78 NTD-PL does not primarily compete by changing the rank tuple, replacing the factor topology, or  
 79 moving to another tensor-network format. Instead, it keeps the Tucker backbone and multilinear  
 80 ranks fixed and changes the measurement map on top of the Tucker latent tensor.

81 **Nonlinear tensor factorization and neural/kernel decoders.** Nonlinear tensor models include  
 82 generalized likelihood and link formulations, Bayesian and kernelized tensor factorizations, Gaussian-  
 83 process tensor models, and neural tensor decoders [Hong et al., 2020, Schein et al., 2016, Xu et al.,

2011, Zhe et al., 2016, Liu et al., 2017, Pan et al., 2020, Saragadam et al., 2024, Varolgunes et al., 2023, Ahn et al., 2024, Luo et al., 2024]. Related imaging methods also learn linear or nonlinear transforms under low-rank tensor priors [Luo et al., 2022, Wang et al., 2024, Xu et al., 2024]. These methods can be more flexible than a scalar polynomial link, but this flexibility often comes with heavier inference, additional hyperparameters, or a less transparent finite-dimensional relation to Tucker. NTD-PL is complementary: it deliberately uses a low-dimensional scalar response rather than a general kernel, Gaussian-process, neural decoder, or learned transform, so that the induced effect can be analyzed relative to a fixed Tucker backbone.

**Polynomial feature lifts and high-order interactions.** Polynomial and Volterra-style expansions are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Related matrix and tensor methods use polynomial feature maps, quadratic factorizations, or explicit high-order interactions to enlarge a linear model [Zhai et al., 2024, Ruegamer, 2024, Liu et al., 2015, Deng et al., 2016]. Such constructions can make nonlinear interactions explicit, but untied high-order parameterizations can increase parameter dimension and contraction cost. NTD-PL does not introduce an independent high-order tensor parameter. Its polynomial powers are powers of one learned Tucker latent tensor, inducing parameter-tied high-order interactions and Veronese-style backbone-tied rank lifts.

**Positioning.** Existing nonlinear tensor models either use broad nonlinear decoders, modify likelihoods or kernels, or explicitly parameterize high-order interactions. What is less developed is a Tucker-preserving, low-dimensional, analyzable polynomial link that yields a parameter-tied rank lift and supports masked learning. NTD-PL trades general nonlinear flexibility for structure, interpretability, and a direct finite-dimensional rank-lift view. Its expected benefit is therefore scoped to settings where matched-rank Tucker leaves a coherent nonlinear residual explained by a scalar response.

## 3 Model

### 3.1 Model and Measurement Assumption

Let  $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$  be a measured tensor and let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad \mathcal{X} \in \mathbb{R}^{R_1 \times \dots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \quad (1)$$

Our measurement assumption is that the measured tensor is generated from this low-rank Tucker latent tensor through a scalar response, up to noise. NTD-PL models this response by a polynomial:

$$Y_{\mathbf{i}} = f_{\beta}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \quad f_{\beta}(s) = \sum_{p=0}^P \beta_p s^p, \quad \mathbf{i} = (i_1, \dots, i_N). \quad (2)$$

Equivalently, the tensor prediction is

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}, \quad \beta \in \mathbb{R}^{P+1}. \quad (3)$$

The constant term  $\beta_0$  captures affine response shifts, the linear term  $\beta_1 \mathcal{S}$  is the usual Tucker channel, and terms with  $p \geq 2$  activate nonlinear channels on the same latent tensor. Standard Tucker is recovered by setting  $\beta_1 = 1$  and  $\beta_p = 0$  for  $p \neq 1$ . Thus NTD-PL changes the measurement map while leaving the Tucker backbone and its multilinear ranks fixed.

### 3.2 Backbone-Tied High-Order Interactions

Although  $f_{\beta}$  is scalar, the term  $\mathcal{S}^{\odot p}$  is not merely an independent output calibration. We arrive at it by first defining an explicit  $p$ -fold Tucker extension: instead of mapping one latent index per mode, the mode- $n$  projection is a higher-order tensor  $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$  that couples  $p$  latent indices. The case  $p = 1$  is ordinary Tucker, while  $p = 2$  gives a quadratic Tucker-style interaction between two latent Tucker tuples. Appendix A.1 gives the formal construction and tensor-network view.

The explicit construction is a reference model, not the parameterization we train. Its projection tensors grow as  $I_n R_n^p$ , and each degree would add a new set of mode projections. NTD-PL keeps the

125 same high-order interaction pattern by tying the  $p$ -fold projection to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}.$$

126 Under this tying, the degree- $p$  channel is exactly  $S_1^p$ . Thus it still couples  $p$  latent Tucker index tuples,  
 127 but all copies share the same mode factors. Across degrees, the only new free parameters are the  
 128 scalar coefficients  $\{\beta_p\}$ , which mix these tied channels.

129 The latent tensor  $\mathcal{S}$  and the coefficients  $\beta$  are fitted jointly, so the low-rank backbone is optimized  
 130 together with the polynomial channels. Section 4 analyzes the induced rank expansion and the  
 131 resulting separation from standard Tucker.

### 132 3.3 Masked Least-Squares Objective

133 Let  $\Omega \in \{0, 1\}^{I_1 \times \dots \times I_N}$  denote the observed-entry mask and let  $|\Omega| = \sum_i \Omega_i$ . We estimate the core,  
 134 factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^N \frac{\lambda_n}{2} \|\mathbf{A}^{(n)}\|_F^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2. \quad (4)$$

135 The same formulation covers full reconstruction by taking  $\Omega \equiv \mathbf{1}$  and tensor completion by restricting  
 136 the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the  
 137 low-dimensional polynomial link.

## 138 4 Theory

139 The theory addresses the rank-lift view raised by the model definition. A scalar link has few free  
 140 parameters, but its powers can induce ordinary Tucker tensors with larger multilinear ranks. We  
 141 first make this induced structure explicit, then give a separation from matched-rank Tucker and an  
 142 approximation route for nonlinear responses. Proofs of all results in this section are provided in  
 143 Appendix B.

### 144 4.1 Rank Expansion and Separation

145 **Proposition 1** (Shared-backbone rank expansion). *Let  $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$  have Tucker rank*  
 146 *at most  $(R_1, \dots, R_N)$ . For every integer  $p \geq 1$ ,  $\mathcal{S}^{\odot p}$  admits a Tucker representation with mode ranks*  
 147 *at most*

$$\left( \binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \quad (5)$$

148 *More explicitly, the ordinary Tucker factors of  $\mathcal{S}^{\odot p}$  can be chosen as Veronese lifts of the original*  
 149 *factors. For  $\mathbf{A} \in \mathbb{R}^{I \times R}$ , define  $\mathbf{A}^{(p)} \in \mathbb{R}^{I \times D}$ ,  $D = \binom{R+p-1}{p}$ , by*

$$A_{i,\alpha}^{(p)} = \prod_{r=1}^R A_{i,r}^{\alpha_r}, \quad (6)$$

150 *where  $\alpha \in \mathbb{N}^R$  ranges over  $|\alpha| = p$ . Then  $\mathcal{S}^{\odot p}$  factors through  $\mathbf{A}^{(1)(p)}, \dots, \mathbf{A}^{(N)(p)}$ .*

151 This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree- $p$   
 152 channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same  
 153 backbone factors. The link therefore changes rank structure through tied features, not through an  
 154 independent high-order parameter block.

155 **Theorem 1** (Tensorial separation from standard Tucker). *Let  $N \geq 3$ ,  $R \geq 2$ ,  $p \geq 2$ , and suppose*  
 156  *$I_n \geq D$  for every  $n$ , where  $D = \binom{R+p-1}{p}$ . Then there exists an order- $N$  tensor  $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$  that*  
 157 *is representable by a degree- $p$  NTD-PL model with Tucker backbone rank  $(R, \dots, R)$ , but whose*  
 158 *exact standard Tucker rank is*

$$(D, \dots, D). \quad (7)$$

159 *Equivalently, a degree- $p$  polynomial link can generate tensors whose multilinear ranks are*  
 160  *$(\binom{R+p-1}{p}, \dots, \binom{R+p-1}{p})$  from the same Tucker backbone of rank  $(R, \dots, R)$ .*

The proof in Appendix B.3 uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker tensors: it identifies a structured high-rank subset generated by a low-rank backbone.

## 4.2 Approximation Under Nonlinear Measurement Responses

**Theorem 2** (Polynomial generation). *Suppose  $\mathcal{Y} = g(\mathcal{S}^*)$ , where  $\mathcal{S}^*$  has Tucker rank at most  $(R_1, \dots, R_N)$  and  $g$  is a polynomial of degree at most  $P$ . Then degree- $P$  NTD-PL with the same Tucker rank can represent  $\mathcal{Y}$  exactly.*

**Theorem 3** (Analytic link approximation). *Suppose  $\mathcal{Y} = g(\mathcal{S}^*)$ , where  $\mathcal{S}^*$  has Tucker rank at most  $(R_1, \dots, R_N)$  and  $\|\mathcal{S}^*\|_\infty \leq B$ . If  $g$  is analytic on the complex disk  $D(0, \rho)$  with  $B < \rho$ , and  $M_\rho = \sup_{|z|=\rho} |g(z)|$ , then there exists a degree- $P$  NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \|\mathcal{Y} - \hat{\mathcal{Y}}\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \quad M = \prod_{n=1}^N I_n. \quad (8)$$

Thus the same finite link that induces a tied rank lift also gives a direct approximation route for smooth responses on a bounded latent range. The statement uses the simple Taylor truncation bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

Together, the results clarify the scope of NTD-PL. Polynomial links can approximate nonlinear responses while keeping the same latent Tucker rank, and their powers induce Veronese-lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without introducing untied high-order tensor parameters.

## 5 Optimization

We optimize the masked objective in Section 3.3 by alternating between the scalar link and the Tucker backbone. The nonlinear part is low-dimensional once the latent Tucker tensor is fixed, while the backbone update remains a standard differentiable Tucker fitting step through the current link. We call the two stages *Link Refresh* and *Linked-Backbone Descent*.

**Link Refresh.** Fix the current latent tensor  $\mathcal{S} = [\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]$ . For active degree  $Q \leq P$ , updating the polynomial coefficients reduces to a small masked ridge regression,

$$\min_{\beta_{0:Q}} \frac{1}{2|\Omega|} \|\mathcal{Y}_\Omega - \Phi_{\Omega,Q}(\mathcal{S})\beta_{0:Q}\|_2^2 + \frac{\lambda_\beta}{2} \|\beta_{0:Q}\|_2^2, \quad (9)$$

where  $\Phi_{\Omega,Q}(\mathcal{S})$  contains the observed powers  $[1, s, \dots, s^Q]$ . This step is cheap because it solves only a  $(Q+1) \times (Q+1)$  linear system. It also gives an interpretable view of the link: once the backbone exposes scalar latent values over the tensor, the refresh estimates the best scalar response on the observed set.

**Linked-Backbone Descent.** With  $\beta$  fixed, the Tucker parameters are updated through the current nonlinear measurement map. The masked residual is first propagated to the latent tensor as

$$T = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - Y) \odot f'_\beta(\mathcal{S}), \quad (10)$$

and then backpropagated through the Tucker reconstruction to update  $\mathcal{X}$  and  $\{\mathbf{A}^{(n)}\}$ . In implementation we use Adam for these first-order steps, but the update itself is just standard Tucker reverse contractions composed with the current link. The backbone is therefore not calibrated after fitting; it is jointly reshaped under the nonlinear link.

**Stabilization.** Directly activating all polynomial degrees from the start is brittle, because higher powers of  $\mathcal{S}$  amplify scale ambiguity. We therefore begin from the Tucker-compatible degree-one model and activate higher degrees gradually. When a new degree is switched on, its coefficient is initialized at zero, so the model expands from the current lower-degree solution without changing the prediction abruptly. We also normalize Tucker factor scales and absorb the corresponding

scales into the core, which fixes a Tucker gauge without changing  $S$ . For the coefficient solve, the implementation may regress on a centered and scaled latent variable and then map the solution back to the original power basis. Appendix Figure 6 shows representative training curves where activating the relevant higher degrees produces systematic loss drops. Full gradients, stabilization details, costs analysis and pseudocode are given in Appendix A.2.

**Masked uniform convergence.** The masked objective also admits a standard sampling interpretation. Let  $m = |\Omega|$  and let  $\Theta \subset \mathbb{R}^d$  be a bounded NTD-PL parameter set, where  $d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P + 1)$ . If the per-sample losses are bounded by  $0 \leq \ell_i(\theta) \leq B_\ell$  and  $G$ -Lipschitz in  $\theta$ , then with probability at least  $1 - \delta$ ,

$$\sup_{\theta \in \Theta} |\mathcal{L}(\theta) - \hat{\mathcal{L}}_m(\theta)| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}, \quad (11)$$

for every  $0 < \varepsilon \leq B_\Theta$ . This is a standard uniform convergence bound, not a sharp minimax result, but it clarifies that masked training controls the full-tensor risk with a complexity term determined by the tied NTD-PL parameter count. The proof is given in Appendix B.4.

## 6 Experiments

The experiments address three mechanism questions. First, under controlled nonlinear residuals, does NTD-PL behave as intended: remaining close to Tucker in the linear regime while gaining when the residual is link-like and the active polynomial degree is expressive enough? Second, on real hyperspectral tensors, does the polynomial link provide a compact fixed-backbone enhancement that improves both representation and held-out completion, rather than merely spending more parameters? Third, can a frozen-backbone Link Refresh diagnostic identify, before full NTD-PL training, which tensor units contain residual structure that a scalar link can exploit? Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. We report  $RMSE$  and  $SAM$  for reconstruction, and  $RMSE^*$  and  $SAM^*$  only on held-out entries for completion. NTD-PL is compared mainly against same-rank Tucker, except in the rank-lift ablation.

### 6.1 Controlled Nonlinear Residuals

These synthetic experiments are mechanism checks rather than generic benchmarks. We generate measured tensors as

$$Y_\alpha = \sqrt{1 - \alpha} S^* + \sqrt{\alpha} \tilde{R}^*,$$

where  $S^*$  is a low-rank Tucker tensor and  $\tilde{R}^*$  is the normalized component of a scalar nonlinear response orthogonal to  $S^*$ . Thus  $\alpha$  controls nonlinear residual energy, not just the raw amplitude of a nonlinearity. Figure 1 shows the expected transition: NTD-PL is Tucker-like when the residual is nearly linear and separates as nonlinear residual energy increases. Tucker can lag behind CP, but NTD-PL then pulls ahead systematically. Figure 2 shows the complementary degree effect: gains emerge when the active polynomial degree can match or approximate the response complexity. Degree continuation and coefficient-contribution details are deferred to Appendix C.1, where Table 4 shows the linear consistency negative control.

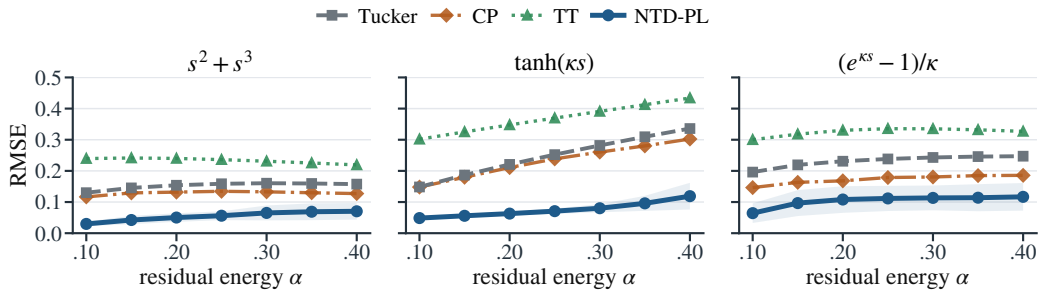


Figure 1: Controlled nonlinear residual sweep. NTD-PL stays Tucker-like near the linear regime and separates as  $\alpha$ , the nonlinear residual energy, increases. Lower RMSE is better.



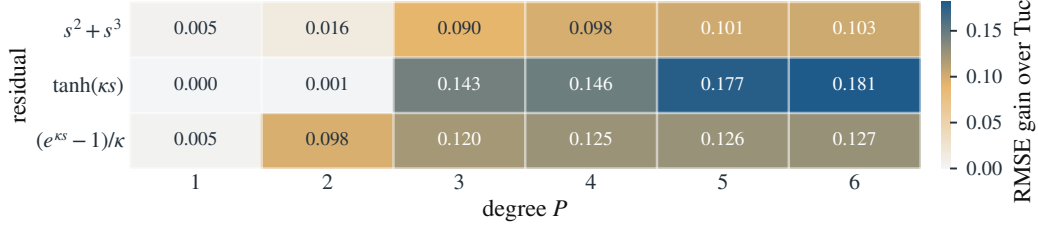


Figure 2: Degree-dependent gains at  $\alpha = 0.25$ . Cell values are Tucker minus NTD-PL RMSE; gains appear when the active degree is sufficient for the response complexity.

Taken together, these controlled experiments answer the first question. NTD-PL does not obtain gains simply by adding a polynomial link: it stays close to Tucker when the source is linear or nearly linear, and its advantage appears when the residual is both nonlinear and expressible by the active degrees. The synthetic results therefore isolate the intended mechanism before moving to real tensors.

## 6.2 Tucker Enhancement on Real Hyperspectral Tensors

This section tests the same fixed-rank enhancement on real hyperspectral tensors, in both full reconstruction and held-out completion. We use CAVE [Columbia CAVE Laboratory, n.d., Yasuma et al., 2010], whose multiple real scenes provide a compact spectral-spatial mechanism check.

**Representation enhancement.** Full reconstruction checks representation capacity rather than generalization. Figure 3 shows that, at matched Tucker backbone rank and similar parameter budget, NTD-PL reduces RMSE and SAM. The gap shrinks as Tucker rank grows, while Table 1 shows that Tucker needs substantially more rank and parameters to match NTD-PL RMSE. NTD-PL also usually keeps a clearer SAM advantage.

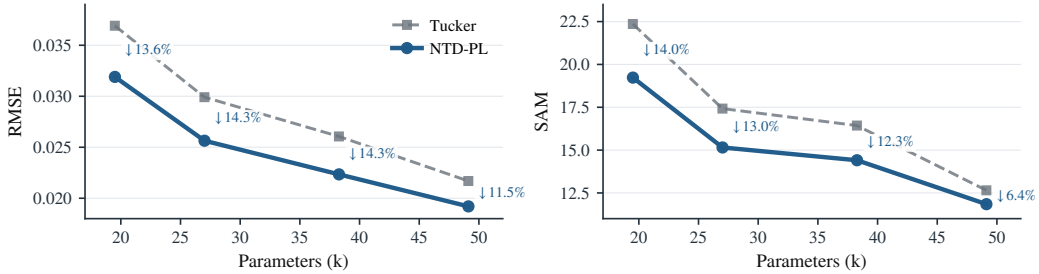


Figure 3: CAVE full reconstruction. At matched backbone rank, NTD-PL reduces RMSE and SAM; annotations show paired gains over Tucker. The gap narrows with Tucker rank, consistent with fixed-rank enhancement. Points average 15 scenes.

Table 1: Rank-lifted Tucker comparison on CAVE reconstruction. For each NTD-PL backbone rank, we report the first higher-rank Tucker model matching NTD-PL RMSE and its extra parameters.

| NTD-PL reference |        |        |              | First Tucker matching RMSE |        |               |       | Extra params |
|------------------|--------|--------|--------------|----------------------------|--------|---------------|-------|--------------|
| Rank             | Params | RMSE↓  | SAM↓         | Rank                       | Params | RMSE↓         | SAM↓  |              |
| (18, 18, 3)      | 20k    | 0.0319 | <b>19.23</b> | (28, 28, 3)                | 31k    | <b>0.0316</b> | 21.41 | +59.5%       |
| (24, 24, 4)      | 27k    | 0.0256 | <b>15.15</b> | (35, 35, 4)                | 41k    | <b>0.0254</b> | 16.26 | +51.3%       |
| (33, 33, 4)      | 38k    | 0.0223 | <b>14.41</b> | (49, 49, 4)                | 60k    | <b>0.0222</b> | 15.28 | +56.5%       |

**Held-out enhancement.** Held-out completion tests transfer to unseen entries. Table 2 reports random-mask CAVE completion, with metrics computed only on missing entries. At the same rank budget, NTD-PL gives consistent RMSE\* gains and usually improves SAM\*; paired sign and Wilcoxon tests in Appendix C.2 show the same scene-level pattern.

These CAVE results answer the second question. The polynomial link improves matched-rank Tucker not only as a representation-capacity check, but also on held-out entries under random missingness. The rank-lift comparison further shows that the gain is not merely a parameter-count effect: at the

Table 2: Held-out CAVE completion at rank (33, 33, 4) under random masks. Metrics use missing entries only; gains are paired improvements over matched-rank Tucker.

| Missing rate | RMSE*↓          |                        |       | SAM*↓        |                     |       |
|--------------|-----------------|------------------------|-------|--------------|---------------------|-------|
|              | Tucker          | NTD-PL                 | Gain  | Tucker       | NTD-PL              | Gain  |
| 0.1          | 0.0263 ± 0.0179 | <b>0.0229 ± 0.0158</b> | 12.9% | 11.82 ± 5.33 | <b>8.36 ± 3.38</b>  | 29.3% |
| 0.3          | 0.0263 ± 0.0179 | <b>0.0230 ± 0.0158</b> | 12.5% | 14.75 ± 6.33 | <b>12.47 ± 5.06</b> | 15.5% |
| 0.5          | 0.0264 ± 0.0179 | <b>0.0230 ± 0.0158</b> | 12.9% | 15.56 ± 6.85 | <b>13.48 ± 5.75</b> | 13.4% |
| 0.7          | 0.0266 ± 0.0180 | <b>0.0219 ± 0.0128</b> | 17.7% | 16.02 ± 7.21 | <b>13.79 ± 5.43</b> | 13.9% |

same Tucker backbone rank, the tied link supplies a compact nonlinear enhancement that higher-rank Tucker can only match with substantially more ordinary multilinear capacity.

### 6.3 Diagnosing Nonlinear Low-Rank Structure Across Data

NTD-PL is meant to be diagnosable rather than uniformly beneficial: it should help when matched-rank Tucker leaves residual structure that a scalar link can reduce. We use  $D_{\text{link}}$  as a cheap frozen-backbone probe: fit Tucker, apply one Link Refresh step from Section 5, and record how much Tucker residual energy this one-step link removes on the evaluation entries. Appendix C.3 gives the exact formula.

For CAVE completion, Table 3 groups scenes by  $D_{\text{link}}$  tercile. Higher scores correspond to larger held-out gains across missing rates, with Spearman correlations around 0.829–0.889; Appendix Figure 7 gives scene-level scatters.

Table 3: CAVE completion diagnostic. Scenes are grouped by  $D_{\text{link}}$  tercile; entries are mean NTD-PL RMSE\* gains over matched-rank Tucker.

| Missing rate | Low $D_{\text{link}}$ | Mid $D_{\text{link}}$ | High $D_{\text{link}}$ | Spearman $\rho_s$ |
|--------------|-----------------------|-----------------------|------------------------|-------------------|
| 0.1          | 0.75%                 | 7.92%                 | 12.63%                 | 0.829             |
| 0.3          | -1.13%                | 9.10%                 | 13.12%                 | 0.871             |
| 0.5          | -1.09%                | 9.16%                 | 13.18%                 | 0.889             |
| 0.7          | -0.62%                | 9.22%                 | 13.07%                 | 0.868             |

We apply the same protocol beyond HSI. Figure 4 plots  $D_{\text{link}}$  against RMSE gain. Across all three families, larger  $D_{\text{link}}$  consistently identifies units with larger NTD-PL gains; object-view tensors show the strongest separation, action video follows the same trend, and natural images retain a positive but lower-yield regime. Appendix Table 7 gives the numeric statistics behind Figure 4.

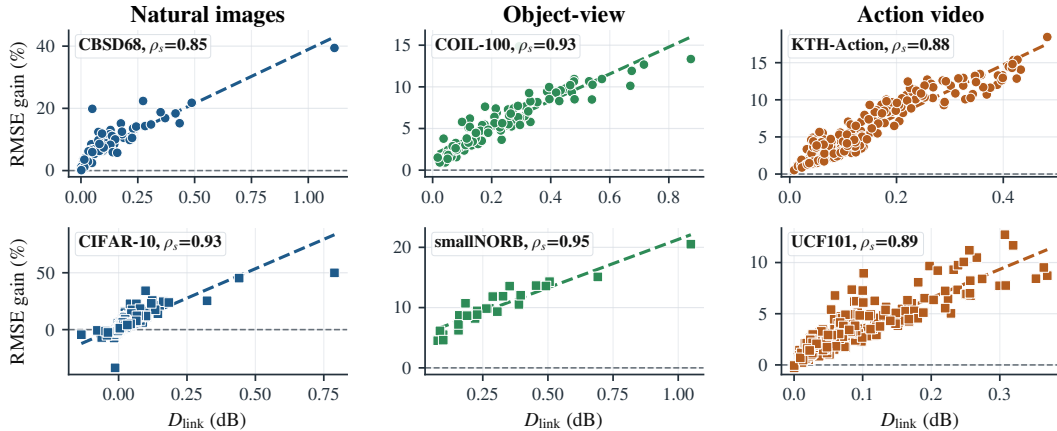


Figure 4: Per-unit cross-domain diagnostic correlation. Each panel plots  $D_{\text{link}}$  against NTD-PL RMSE gain over independently fitted tensor units.

These diagnostic results answer the third question. A single frozen-backbone Link Refresh score is already predictive of where full NTD-PL will help, across both CAVE completion and non-HSI tensor units. This supports the scoped claim of the paper: NTD-PL is useful when matched-rank



273 Tucker leaves residual structure that is coherent with a shared scalar response, and low diagnostic  
274 yield correctly signals cases where the polynomial link should provide little benefit.

## 275 7 Limitations

276 NTD-PL assumes that a common nonlinear response explains a coherent part of the residual left by  
277 matched-rank Tucker. A single scalar link may be less effective when nonlinearities vary strongly  
278 across spatial regions, spectral bands, modes, or samples; when the remaining error is ordinary  
279 multilinear structure rather than a scalar response, increasing Tucker rank remains the more direct  
280 modeling choice.

281 Polynomial degree adds approximation capacity but can amplify Tucker scale ambiguity and con-  
282 ditioning issues in the power basis, motivating the ridge-regularized Link Refresh step, factor  
283 normalization, and degree continuation. The joint problem also remains nonconvex: Link Refresh is  
284 a small ridge regression with fixed latent tensor, but the backbone and link are learned jointly, and we  
285 do not claim global optimality.

286 Our strongest empirical evidence is from controlled nonlinear residuals and hyperspectral reconstruc-  
287 tion and completion. Broader validation on large sparse tensors, temporal or multimodal tensors, and  
288 domain-varying nonlinear responses is left for future work.

## 289 8 Conclusion

290 We asked whether a Tucker model can keep its chosen backbone ranks fixed while learning a  
291 structured response that explains coherent nonlinear residuals left by matched-rank Tucker. NTD-PL  
292 gives a positive but scoped answer. It places a polynomial link on a shared Tucker latent tensor,  
293 so each higher-degree channel is not an independent high-rank correction but a Veronese-lifted,  
294 parameter-tied rank lift of the same backbone. In this way, the model preserves the core-factor  
295 structure and rank budget of Tucker while adding only a small number of scalar link coefficients.

296 The analysis and optimizer are built around this shared-backbone view. The theory shows how  
297 polynomial powers induce tied rank expansion, separates the resulting model class from matched-  
298 rank Tucker, and provides approximation guarantees for nonlinear measurement responses. The  
299 optimization procedure alternates a small Link Refresh ridge solve with Linked-Backbone Descent  
300 and degree continuation, making the link a jointly learned part of the representation rather than a  
301 post-hoc calibration layer.

302 The experiments identify the regime in which this additional structure is useful. Controlled residual  
303 experiments show that NTD-PL stays close to Tucker in linear or nearly linear settings and gains  
304 when the residual is nonlinear and expressible by the active polynomial degrees. CAVE reconstruction  
305 and completion show that the same fixed-backbone enhancement improves both representation and  
306 held-out prediction, while the rank-lift ablation shows that the gain is not merely a parameter-count  
307 effect. Finally, the Link Refresh diagnostic predicts where full NTD-PL helps across HSI and non-HSI  
308 tensor units. These results position NTD-PL not as a universal nonlinear tensor decoder, but as a  
309 compact Tucker enhancement for tensors whose matched-rank residuals are coherent with a shared  
310 scalar response. Future work may extend this idea to adaptive links, mode- or region-dependent  
311 responses, and scalable sparse implementations.

## 312 References

- 313 Dawon Ahn, Uday Singh Saini, Evangelos E. Papalexakis, and Ali Payani. Neural additive tensor  
314 decomposition for sparse tensors. In *Proceedings of the 33rd ACM International Conference on*  
315 *Information and Knowledge Management*, pages 14–23, 2024. doi: 10.1145/3627673.3679833.
- 316 José M. Bioucas-Dias, Antonio Plaza, Nicolas Dobigeon, Mario Parente, Qian Du, Paul Gader,  
317 and Jocelyn Chanussot. Hyperspectral unmixing overview: Geometrical, statistical, and sparse  
318 regression-based approaches. *IEEE Journal of Selected Topics in Applied Earth Observations and*  
319 *Remote Sensing*, 5(2):354–379, 2012. doi: 10.1109/JSTARS.2012.2194696.

Stephen Boyd and Leon O. Chua. Fading memory and the problem of approximating nonlinear operators with volterra series. *IEEE Transactions on Circuits and Systems*, 32(11):1150–1161, 1985. doi: 10.1109/TCS.1985.1085649.

Columbia CAVE Laboratory. Multispectral image database. <https://cave.cs.columbia.edu/repository/Multispectral>, n.d. Accessed: 2026-04-13.

Lieven De Lathauwer, Bart De Moor, and Joos Vandewalle. A multilinear singular value decomposition. *SIAM Journal on Matrix Analysis and Applications*, 21(4):1253–1278, 2000. doi: 10.1137/S0895479896305696.

Paul E. Debevec and Jitendra Malik. Recovering high dynamic range radiance maps from photographs. In *Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques*, SIGGRAPH ’97, pages 369–378, 1997. doi: 10.1145/258734.258884.

Jian Deng, Haotian Liu, Kim Batselier, Yu-Kwong Kwok, and Ngai Wong. Storm: a nonlinear model order reduction method via symmetric tensor decomposition. In *2016 21st Asia and South Pacific design automation conference (ASP-DAC)*, pages 557–562. IEEE, 2016.

Michael D. Grossberg and Shree K. Nayar. Modeling the space of camera response functions. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(10):1272–1282, 2004. doi: 10.1109/TPAMI.2004.88.

Rob Heylen, Mario Parente, and Paul Gader. A review of nonlinear hyperspectral unmixing methods. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 7(6):1844–1868, 2014. doi: 10.1109/JSTARS.2014.2320576.

David Hong, Tamara G Kolda, and Jed A Duersch. Generalized canonical polyadic tensor decomposition. *SIAM review*, 62(1):133–163, 2020.

Tamara G. Kolda and Brett W. Bader. Tensor decompositions and applications. *SIAM Review*, 51(3): 455–500, 2009. doi: 10.1137/07070111X.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL <https://www.cs.toronto.edu/~kriz/cifar.html>. Dataset page accessed 2026-05-07.

Yann LeCun, Fu Jie Huang, and Léon Bottou. Learning methods for generic object recognition with invariance to pose and lighting. In *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, volume 2, pages 97–104, 2004. doi: 10.1109/CVPR.2004.1315150. URL <https://cs.nyu.edu/~ylclab/data/norb-v1.0-small/>.

Allen Liu and Ankur Moitra. Tensor completion made practical. In *Advances in Neural Information Processing Systems*, volume 33, pages 18905–18916, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/dab1263d1e6a88c9ba5e7e294def5e8b-Abstract.html>.

Bin Liu, Lirong He, Shandian Zhe, Yingmin Li, and Zenglin Xu. Deepcp: Flexible nonlinear tensor decomposition. In *NeurIPS Workshops*, volume 3, 2017.

Haotian Liu, Luca Daniel, and Ngai Wong. Model reduction and simulation of nonlinear circuits via tensor decomposition. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 34(7):1059–1069, 2015.

Ji Liu, Przemyslaw Musialski, Peter Wonka, and Jieping Ye. Tensor completion for estimating missing values in visual data. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(1):208–220, 2013. doi: 10.1109/TPAMI.2012.39.

Yi-Si Luo, Xi-Le Zhao, Tai-Xiang Jiang, Yi Chang, Michael K. Ng, and Chao Li. Self-supervised nonlinear transform-based tensor nuclear norm for multi-dimensional image recovery. *IEEE Transactions on Image Processing*, 31:3793–3808, 2022. doi: 10.1109/TIP.2022.3176220.

Yisi Luo, Xile Zhao, Zhemin Li, Michael K. Ng, and Deyu Meng. Low-rank tensor function representation for multi-dimensional data recovery. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3351–3369, 2024. doi: 10.1109/TPAMI.2023.3341688.

David Martin, Charless Fowlkes, Doron Tal, and Jitendra Malik. A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 416–423, 2001. URL <https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/>.

Sameer A. Nene, Shree K. Nayar, and Hiroshi Murase. Columbia Object Image Library (COIL-100). Technical Report CUCS-006-96, Columbia University, 1996. URL <https://cave.cs.columbia.edu/repository/COIL-100>. Dataset page accessed 2026-05-07.

Ivan V. Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33(5): 2295–2317, 2011.

Zhimeng Pan, Zheng Wang, and Shandian Zhe. Streaming nonlinear bayesian tensor decomposition. In *Conference on Uncertainty in Artificial Intelligence*, pages 490–499. PMLR, 2020.

David Ruegamer. Scalable higher-order tensor product spline models. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 1–9. PMLR, 2024. URL <https://proceedings.mlr.press/v238/ruegamer24a.html>.

Vishwanath Saragadam, Randall Balestrieri, Ashok Veeraraghavan, and Richard G Baraniuk. Deeptensor: Low-rank tensor decomposition with deep network priors. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10337–10348, 2024.

Aaron Schein, Mingyuan Zhou, David Blei, and Hanna Wallach. Bayesian poisson tucker decomposition for learning the structure of international relations. In *International Conference on Machine Learning*, pages 2810–2819. PMLR, 2016.

Christian Schuldt, Ivan Laptev, and Barbara Caputo. Recognizing human actions: A local SVM approach. In *Proceedings of the 17th International Conference on Pattern Recognition*, volume 3, pages 32–36, 2004. doi: 10.1109/ICPR.2004.1334462. URL <https://www.csc.kth.se/cvap/actions/>.

Nicholas D. Sidiropoulos, Lieven De Lathauwer, Xiao Fu, Kejun Huang, Evangelos E. Papalexakis, and Christos Faloutsos. Tensor decomposition for signal processing and machine learning. *IEEE Transactions on Signal Processing*, 65(13):3551–3582, 2017. doi: 10.1109/TSP.2017.2690524.

Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. UCF101: A dataset of 101 human action classes from videos in the wild. Technical Report CRCV-TR-12-01, University of Central Florida, 2012. URL <https://www.crcv.ucf.edu/data/UCF101.php>. Dataset page and technical report accessed 2026-05-07.

Uras Varolgunes, Dan Zhou, Dantong Yu, and Ajim Uddin. Nmtucker: Non-linear matryoshka tucker decomposition for financial time series imputation. In *Proceedings of the Fourth ACM International Conference on AI in Finance*, pages 516–523, 2023.

Vito Volterra. *Theory of Functionals and of Integral and Integro-Differential Equations*. Dover Publications, 1959.

Jian-Li Wang, Ting-Zhu Huang, Xi-Le Zhao, Tai-Xiang Jiang, and Michael K. Ng. Multi-dimensional visual data completion via low-rank tensor representation under coupled transform. *IEEE Transactions on Image Processing*, 30:3581–3596, 2021. doi: 10.1109/TIP.2021.3062995.

Jian-Li Wang, Ting-Zhu Huang, Xi-Le Zhao, Yi-Si Luo, and Tai-Xiang Jiang. CoNoT: Coupled nonlinear transform-based low-rank tensor representation for multidimensional image completion. *IEEE Transactions on Neural Networks and Learning Systems*, 35(7):8969–8983, 2024. doi: 10.1109/TNNLS.2022.3217198.

Shuang Xu, Jiangjun Peng, Teng-Yu Ji, Xiangyong Cao, Kai Sun, Rongrong Fei, and Deyu Meng. Stacked tucker decomposition with multi-nonlinear products for remote sensing imagery inpainting. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–13, 2024.

- 416 Zenglin Xu, Feng Yan, et al. Infinite tucker decomposition: Nonparametric bayesian models for  
417 multiway data analysis. *arXiv preprint arXiv:1108.6296*, 2011.
- 418 Fumihito Yasuma, Tomoo Mitsunaga, Daisuke Iso, and Shree K. Nayar. Generalized assorted pixel  
419 camera: Postcapture control of resolution, dynamic range, and spectrum. *IEEE Transactions on*  
420 *Image Processing*, 19(9):2241–2253, 2010. doi: 10.1109/TIP.2010.2046811.
- 421 Zheng Zhai, Hengchao Chen, and Qiang Sun. Quadratic matrix factorization with applications  
422 to manifold learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(9):  
423 6384–6401, 2024.
- 424 Qibin Zhao, Guoxu Zhou, Shengli Xie, Liqing Zhang, and Andrzej Cichocki. Tensor ring decomposi-  
425 tion. *arXiv preprint arXiv:1606.05535*, 2016.
- 426 Shandian Zhe, Kai Zhang, Pengyuan Wang, Kuang-chih Lee, Zenglin Xu, Yuan Qi, and Zoubin  
427 Ghahramani. Distributed flexible nonlinear tensor factorization. *Advances in neural information*  
428 *processing systems*, 29, 2016.
- 429 Yu-Bang Zheng, Ting-Zhu Huang, Xi-Le Zhao, Qibin Zhao, and Tai-Xiang Jiang. Fully-connected  
430 tensor network decomposition and its application to higher-order tensor completion. In *Proceedings*  
431 *of the AAAI Conference on Artificial Intelligence*, volume 35, pages 11071–11078, 2021. doi:  
432 10.1609/aaai.v35i12.17321.

## 433 A Additional Model Details

### 434 A.1 Explicit $p$ -Fold Tucker Motivation

435 This section expands the motivation behind the explicit  $p$ -fold Tucker channel in Section 3.2. The  
436 construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map.  
437 In the matrix case, one may move from a linear model  $\mathbf{y} = \mathbf{A}\mathbf{x}$  to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

438 where the second term couples pairs of latent coordinates. The corresponding question for Tucker is:  
439 if standard Tucker is the multilinear low-rank map for multiway data, what is the direct  $p$ -th order  
440 interaction analogue?

441 For a Tucker backbone, an explicit  $p$ -fold interaction can be written as

$$\tilde{Y}_i^{(p)} = \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \left( \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)}. \quad (12)$$

442 Here each mode projection depends on  $p$  latent indices instead of one. For  $p = 1$ , this reduces to the  
443 ordinary Tucker entry formula. For  $p = 2$ , the same core appears twice and the mode projections  
444 couple two latent tuples, giving a quadratic Tucker-style channel. Figure 5 illustrates this progression.

445 The explicit construction is useful as a reference model, but it is not the parameterization used by  
446 NTD-PL. Its mode projections  $\mathcal{A}^{(n,p)}$  have size  $I_n R_n^p$ , so a direct implementation introduces new  
447 high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the  
448 same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}. \quad (13)$$

449 Substituting (13) into (12) gives exactly  $\mathcal{S}^{\odot p}$ , where  $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ . Thus the degree- $p$   
450 NTD-PL channel is a shared-projection version of the explicit  $p$ -fold Tucker prototype.

451 This is the modeling role of the  $p$ -fold Tucker prototype: it defines the high-order Tucker interaction  
452 that NTD-PL compresses. The final model does not train the untied tensors  $\mathcal{A}^{(n,p)}$ ; it keeps a single  
453 Tucker backbone and adds only the polynomial coefficients  $\{\beta_p\}$ .

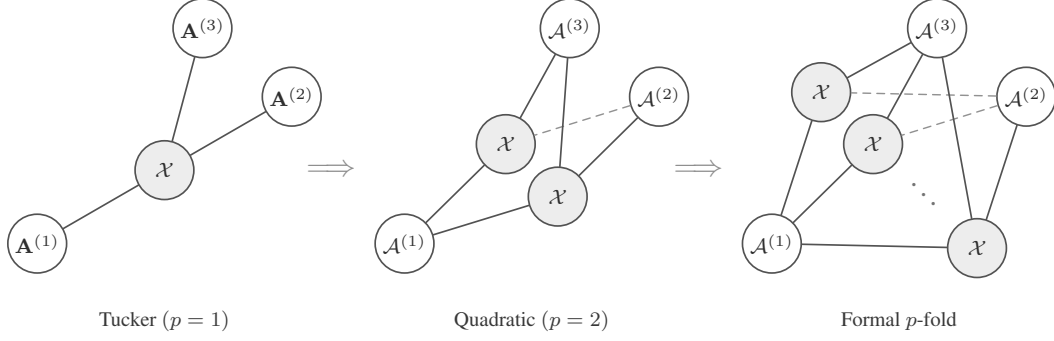


Figure 5: Tensor-network view of a formal  $p$ -fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

## 454 A.2 Optimization Details

455 This section gives the update equations behind the solver summarized in Section 5. The implementa-  
 456 tion follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the  
 457 polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression  
 458 updates of the link coefficients.

459 **Backbone gradients.** For a fixed  $\beta$ , define

$$f'_\beta(\mathcal{S}) = \sum_{p=1}^P p\beta_p \mathcal{S}^{\odot(p-1)}. \quad (14)$$

460 The gradient of the normalized masked data term with respect to the latent tensor  $\mathcal{S}$  is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - \mathcal{Y}) \odot f'_\beta(\mathcal{S}). \quad (15)$$

461 Backpropagating  $\mathbf{T}$  through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \cdots \times_N \mathbf{A}^{(N)\top} + \lambda_{\mathcal{X}} \mathcal{X}. \quad (16)$$

462 For the  $n$ -th factor, let

$$\mathbf{Z}^{(n)} = \left[ \mathcal{X} \times_1 \mathbf{A}^{(1)} \cdots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \cdots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \quad (17)$$

463 Then

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \quad (18)$$

464 The experiments use Adam for these first-order steps. This choice is not part of the model definition;  
 465 any comparable first-order optimizer can use the same gradients.

466 **Link update.** With  $\mathcal{S}$  fixed and active degree  $Q = P_{\text{act}}$ , the coefficient subproblem is the ridge  
 467 regression in (9). Its closed-form solution is

$$\beta_{0:Q}^* = (\Phi_{\Omega,Q}^\top \Phi_{\Omega,Q} + |\Omega| \lambda_\beta \mathbf{I})^{-1} \Phi_{\Omega,Q}^\top \mathbf{y}_\Omega. \quad (19)$$

468 Equivalently, one may absorb  $|\Omega|$  into the ridge parameter and solve the usual unnormalized normal  
 469 equations. If the constant term is disabled, the first column is removed and  $\beta_0$  is fixed at zero. In  
 470 the code, this small system is solved either from an explicit Vandermonde design or from moment  
 471 equations built from cached powers.

472 **Stabilization and continuation.** High powers of  $\mathcal{S}$  can be sensitive to scale. The implementation  
 473 therefore normalizes factor columns after backbone updates and absorbs the corresponding scales  
 474 into the core; this is a gauge fixing of the Tucker parameterization and does not change  $\mathcal{S}$ . For the  
 475 coefficient solve, it may regress on the centered and scaled latent variable  $(S - \mu)/\sigma$ , then convert  
 476 the solution back to the original power basis of  $\mathcal{S}$ .

477 The active degree follows the schedule

$$t_k = \text{round} \left( \frac{kT}{P} \right), \quad k = 1, \dots, P-1.$$

478 When degree  $p+1$  is activated, the previous coefficients and Tucker factors are kept and the new  
479 coefficient  $\beta_{p+1}$  is initialized at zero.

480 **Cost.** The dominant cost is forming the Tucker reconstruction and its reverse contractions on the  
481 observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same  
482 rank, plus  $O(P|\Omega|)$  work to evaluate the scalar polynomial and its derivative. The link update forms  
483 a Vandermonde design or the corresponding moment equations and solves a  $(P+1) \times (P+1)$  linear  
484 system, with direct cost  $O(P^2|\Omega| + P^3)$ . Since  $P$  is small, this cost is negligible compared with the  
485 Tucker contractions.

486 Algorithm 1 gives the full procedural summary.

---

**Algorithm 1** Masked NTD-PL optimization

---

**Require:** observed tensor  $\mathcal{Y}$ , mask  $\Omega$ , rank  $(R_1, \dots, R_N)$ , degree  $P$

- 1: initialize  $\mathcal{X}, \{\mathbf{A}^{(n)}\}$  from Tucker on the masked tensor
  - 2: initialize active degree  $P_{\text{act}} \leftarrow 1$  and  $\beta = (0, 1)$
  - 3: **for**  $t = 1, \dots, T$  **do**
  - 4:   form  $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$
  - 5:   **if** a link update is scheduled **then**
  - 6:     update active coefficients  $\beta_{0:P_{\text{act}}}$  by ridge regression on  $\{(S_i, Y_i) : \Omega_i = 1\}$
  - 7:   **end if**
  - 8:   compute  $f_\beta(\mathcal{S})$  and  $f'_\beta(\mathcal{S})$
  - 9:   take first-order steps on  $\mathcal{X}$  and  $\{\mathbf{A}^{(n)}\}$  using the masked gradient
  - 10:   normalize factor scales and absorb them into the core
  - 11:   increase  $P_{\text{act}}$  according to the continuation schedule, initializing the new coefficient at zero
  - 12: **end for**
  - 13: **return**  $\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta$
- 

## 487 B Additional Proofs

### 488 B.1 Polynomial and Analytic Approximation

489 *Proof of Theorem 2.* Since  $g$  has degree at most  $P$ , write

$$g(s) = \sum_{p=0}^P c_p s^p. \quad (20)$$

490 Take the NTD-PL latent tensor to be the true latent tensor,  $\mathcal{S} = \mathcal{S}^*$ , and set  $\beta_p = c_p$ . Then, entrywise,

$$\hat{\mathcal{Y}}_{i_1, \dots, i_N} = \sum_{p=0}^P \beta_p (\mathcal{S}_{i_1, \dots, i_N}^*)^p = g(\mathcal{S}_{i_1, \dots, i_N}^*) = \mathcal{Y}_{i_1, \dots, i_N}. \quad (21)$$

491 Thus  $\hat{\mathcal{Y}} = \mathcal{Y}$ . □

492 **Lemma 1** (Scalar-to-tensor approximation). *Let  $\|\mathcal{S}^*\|_\infty \leq B$ . For any scalar function  $g$  and any*  
493 *degree- $P$  polynomial  $q_P$ ,*

$$\|g(\mathcal{S}^*) - q_P(\mathcal{S}^*)\|_F \leq \sqrt{\prod_{n=1}^N I_n} \sup_{|s| \leq B} |g(s) - q_P(s)|. \quad (22)$$

494 *Proof.* Every entry of  $\mathcal{S}^*$  lies in  $[-B, B]$ , so every entrywise approximation error is bounded by  
495  $\sup_{|s| \leq B} |g(s) - q_P(s)|$ . Squaring, summing over all entries, and taking the square root gives the  
496 claim. □



497 *Proof of Theorem 3.* Let  $g(s) = \sum_{k=0}^{\infty} a_k s^k$  be the Taylor expansion of  $g$  around zero, and choose  
 498  $q_P(s) = \sum_{k=0}^P a_k s^k$ . By Cauchy's estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \quad (23)$$

499 For  $|s| \leq B < \rho$ ,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^{\infty} |a_k| |s|^k \quad (24)$$

$$\leq M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \quad (25)$$

500 Applying Lemma 1 and taking  $\mathcal{S} = \mathcal{S}^*$ ,  $\beta_k = a_k$  for  $0 \leq k \leq P$ , yields the stated unnormalized  
 501 bound. Dividing by  $\sqrt{M}$ ,  $M = \prod_{n=1}^N I_n$ , gives the normalized form used in Theorem 3.  $\square$

## 502 B.2 Rank Expansion

503 *Proof of Proposition 1.* Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1, \dots, i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}. \quad (26)$$

504 Taking an element-wise  $p$ -th power gives

$$(\mathcal{S}_{i_1, \dots, i_N})^p = \prod_{q=1}^p \left( \sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \prod_{n=1}^N A_{i_n, r_n^{(q)}}^{(n)} \right) \quad (27)$$

$$= \sum_{\gamma_1 \in [R_1]^p} \cdots \sum_{\gamma_N \in [R_N]^p} \left( \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \left( \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} \right), \quad (28)$$

505 where  $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$ . For each mode  $n$ , let  $\alpha_n \in \mathbb{N}^{R_n}$  record the counts of the entries in  $\gamma_n$ ,  
 506 so  $|\alpha_n| = p$ . Then

$$\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} = \prod_{r=1}^{R_n} (A_{i_n, r}^{(n)})^{\alpha_{n,r}} = A_{i_n, \alpha_n}^{(n)\langle p \rangle}. \quad (29)$$

507 Group all ordered tuples  $(\gamma_1, \dots, \gamma_N)$  that produce the same multi-indices  $(\alpha_1, \dots, \alpha_N)$ , and define  
 508 the aggregated core

$$\mathcal{X}_{\alpha_1, \dots, \alpha_N}^{\langle p \rangle} = \sum_{\substack{\gamma_1, \dots, \gamma_N: \\ \text{count}(\gamma_n) = \alpha_n \ \forall n}} \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}}. \quad (30)$$

509 The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = \llbracket \mathcal{X}^{\langle p \rangle}; \mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle} \rrbracket. \quad (31)$$

510 The  $n$ -th Veronese factor has  $\binom{R_n+p-1}{p}$  columns, giving the claimed Tucker rank bound.  $\square$

## 511 B.3 Separation

512 **Lemma 2** (Full-rank Veronese evaluations). *Let  $D = \binom{R+p-1}{p}$ . There exists a matrix  $\mathbf{A} \in \mathbb{R}^{D \times R}$*   
 513 *whose Veronese feature matrix  $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{D \times D}$  has rank  $D$ . Consequently, for  $I \geq D$ , a generic*  
 514 *matrix  $\mathbf{A} \in \mathbb{R}^{I \times R}$  has  $\text{rank}(\mathbf{A}^{\langle p \rangle}) = D$ .*

515 *Proof.* The homogeneous degree- $p$  monomials in  $R$  variables are linearly independent polynomials.  
 516 Hence there exist evaluation points  $x_1, \dots, x_D \in \mathbb{R}^R$  such that the matrix  $(x_i^\alpha)_{i,\alpha}$ ,  $|\alpha| = p$ , is  
 517 nonsingular; otherwise every  $D \times D$  evaluation determinant would vanish, which would imply a  
 518 nontrivial linear dependence among the monomials. Taking these points as the rows of  $\mathbf{A}$  gives the  
 519 first claim.

520 For  $I \geq D$ , choose any  $D$  rows. The determinant of the corresponding  $D \times D$  Veronese submatrix  
 521 is a polynomial in the entries of  $\mathbf{A}$  that is not identically zero by the first claim. Its nonvanishing is  
 522 therefore a generic property, so  $\mathbf{A}^{(p)}$  has full column rank for generic  $\mathbf{A}$ .  $\square$

523 *Proof of Theorem 1.* Let  $D = \binom{R+p-1}{p}$ . Choose generic factor matrices  $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in$   
 524  $\mathbb{R}^{I_n \times R}$ , and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^R a_r^{(1)} \otimes \dots \otimes a_r^{(N)}. \quad (32)$$

525 Its mode- $n$  unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left( \mathbf{A}^{(N)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)} \right)^\top. \quad (33)$$

526 For generic  $\mathbf{A}^{(n)}$ ,  $\mathbf{A}^{(n)}$  has rank  $R$ , and the Khatri-Rao product on the right has rank  $R$ . Thus  
 527  $\text{rank}(\mathcal{S}_{(n)}) = R$  for every  $n$ , so  $\mathcal{S}$  has exact Tucker rank  $(R, \dots, R)$ .

528 Set  $\beta_p = 1$  and  $\beta_q = 0$  for  $q \neq p$ . Then  $\mathcal{Y} = \mathcal{S}^{\odot p}$  is a valid degree- $p$  NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1, \dots, i_N} = \left( \sum_{r=1}^R \prod_{n=1}^N A_{i_n, r}^{(n)} \right)^p \quad (34)$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^N \prod_{r=1}^R \left( A_{i_n, r}^{(n)} \right)^{\alpha_r}, \quad (35)$$

529 where  $c_\alpha = p! / (\alpha_1! \dots \alpha_R!) > 0$ . Let  $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$ . By Lemma 2, each  $\mathbf{B}^{(n)}$  has full  
 530 column rank  $D$  for generic  $\mathbf{A}^{(n)}$ . The mode- $n$  unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \text{diag}(c) \left( \mathbf{B}^{(N)} \odot \dots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \dots \odot \mathbf{B}^{(1)} \right)^\top. \quad (36)$$

531 The diagonal matrix  $\text{diag}(c)$  is nonsingular. Since a Khatri-Rao product of full-column-rank matrices  
 532 is full column rank, the right factor has rank  $D$ . Therefore  $\text{rank}(\mathcal{Y}_{(n)}) = D$ . This holds for every  
 533 mode  $n$ , and the exact Tucker rank of  $\mathcal{Y}$  is  $(D, \dots, D)$ .  $\square$

#### 534 B.4 Masked Generalization

535 **Theorem 4** (Masked-entry uniform convergence). *Let  $\Theta \subset \mathbb{R}^d$  be a bounded NTD-PL parameter set*  
 536 *with*

$$d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P + 1).$$

537 For  $\theta \in \Theta$ , define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \quad M = \prod_{n=1}^N I_n,$$

538 and the masked empirical risk

$$\hat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^m \ell_{\mathbf{i}_t}(\theta),$$

where  $\mathbf{i}_t$  are sampled uniformly from tensor entries. Assume  $0 \leq \ell_{\mathbf{i}}(\theta) \leq B_\ell$  and  $\ell_{\mathbf{i}}(\theta)$  is  $G$ -Lipschitz in  $\theta$ , uniformly over  $\mathbf{i}$ . If  $\Theta$  is contained in a Euclidean ball of radius  $B_\Theta$ , then with probability at least  $1 - \delta$ ,

$$\sup_{\theta \in \Theta} |\mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta)| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

for every  $0 < \varepsilon \leq B_\Theta$ .

*Proof of Theorem 4.* Fix  $0 < \varepsilon \leq B_\Theta$ , and let  $\mathcal{N}_\varepsilon$  be an  $\varepsilon$ -net of  $\Theta$  in Euclidean norm. Since  $\Theta$  lies in a  $d$ -dimensional ball of radius  $B_\Theta$ , it admits a net with

$$|\mathcal{N}_\varepsilon| \leq \left( \frac{3B_\Theta}{\varepsilon} \right)^d. \quad (37)$$

For any fixed  $\theta_0 \in \mathcal{N}_\varepsilon$ , the random variables  $\ell_{\mathbf{i}_t}(\theta_0)$  are independent and bounded in  $[0, B_\ell]$ . Hoeffding's inequality gives

$$\Pr \left( |\mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0)| \geq u \right) \leq 2 \exp \left( -\frac{2mu^2}{B_\ell^2} \right). \quad (38)$$

Taking a union bound over  $\mathcal{N}_\varepsilon$ , with probability at least  $1 - \delta$ ,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} |\mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0)| \leq B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \quad (39)$$

Substituting the net-size bound yields the stated stochastic term.

Now take any  $\theta \in \Theta$ , and choose  $\theta_0 \in \mathcal{N}_\varepsilon$  with  $\|\theta - \theta_0\|_2 \leq \varepsilon$ . By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \leq G\varepsilon, \quad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \leq G\varepsilon. \quad (40)$$

Therefore

$$|\mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta)| \leq |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + |\mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0)| \quad (41)$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \quad (42)$$

$$\leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \quad (43)$$

Taking the supremum over  $\theta \in \Theta$  completes the proof.  $\square$

**Remark 1.** The Lipschitz and bounded-loss assumptions are standard for compact parametric classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial coefficients are bounded and the polynomial degree  $P$  is finite. The resulting Lipschitz constant may depend polynomially on these bounds and on  $P$ , but the covering dimension remains the number of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker expansion.

## C Additional Experiments

### C.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 4 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting  $Y = g(S^*)$ . Let  $S^*$  be the linear low-rank Tucker backbone and let  $g$  be one of the candidate scalar responses. We form

$$R^* = g(S^*) - \frac{\langle g(S^*), S^* \rangle}{\|S^*\|_F^2} S^*, \quad (44)$$

Table 4: Linear consistency on synthetic Tucker tensors. In the linear negative control, effective  $p > 1$  contributions remain small except for the intended affine offset case.

| Method                 | bias = 0      |         | bias = 0.5    |         |
|------------------------|---------------|---------|---------------|---------|
|                        | RMSE↓         | $p > 1$ | RMSE↓         | $p > 1$ |
| Tucker                 | <b>0.0114</b> | –       | 0.0647        | –       |
| NTD-PL, $\beta_0 = 0$  | 0.0116        | 4.19%   | 0.0522        | 39.17%  |
| NTD-PL, $\beta_0$ free | 0.0117        | 3.19%   | <b>0.0118</b> | 2.98%   |

and, when  $\|R^*\|_F > 0$ , normalize it as

$$\tilde{R}^* = \frac{\|S^*\|_F}{\|R^*\|_F} R^*. \quad (45)$$

The observed tensor is then

$$Y_\alpha = \sqrt{1 - \alpha} S^* + \sqrt{\alpha} \tilde{R}^*, \quad 0 \leq \alpha \leq 1. \quad (46)$$

By construction,  $\langle S^*, \tilde{R}^* \rangle = 0$  and  $\|\tilde{R}^*\|_F = \|S^*\|_F$ , so  $\alpha$  is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are  $g(s) = s^2 + s^3$ ,  $g(s) = (\exp(\kappa s) - 1)/\kappa$ , and  $g(s) = \tanh(\kappa s)$ , with  $\kappa$  set from the input range.

The main text reports the nonlinear alpha sweep and degree sweep. Figure 6 shows that degree continuation activates higher powers with systematic loss drops at relevant degrees. Table 5 complements those figures with a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

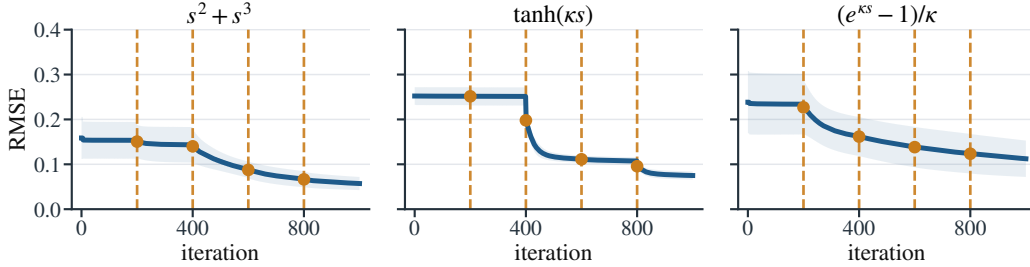


Figure 6: NTD-PL training curves for controlled nonlinear mechanism checks at  $\alpha = 0.25$  and  $P = 5$ . Dashed vertical lines mark degree activations, which produce systematic drops when the activated degree matches the response structure.

Table 5: Effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

| Link                        | Effective contribution (%) |             |         |             |         |         |         |
|-----------------------------|----------------------------|-------------|---------|-------------|---------|---------|---------|
|                             | $p = 0$                    | $p = 1$     | $p = 2$ | $p = 3$     | $p = 4$ | $p = 5$ | $p = 6$ |
| $s^2 + s^3$                 | 0.1                        | 24.1        | 9.8     | <b>38.1</b> | 7.3     | 11.5    | 9.1     |
| $(e^{\kappa s} - 1)/\kappa$ | 0.3                        | <b>39.8</b> | 16.8    | 13.2        | 10.1    | 7.7     | 12.1    |
| $\tanh(\kappa s)$           | 0.2                        | 37.9        | 1.3     | <b>30.9</b> | 5.4     | 17.3    | 7.0     |

## C.2 CAVE Completion Significance and Robustness

The main text reports the full-resolution random-missing CAVE completion table at rank  $(33, 33, 4)$ . We do not repeat that table here; instead, Table 6 gives the paired sign and Wilcoxon tests for the same protocol.

Table 6: Paired significance tests for full-resolution CAVE completion at rank (33, 33, 4). Metrics are computed only on missing entries.

| $\rho$ | RMSE |           |                       |                       | SAM  |           |                       |                       |
|--------|------|-----------|-----------------------|-----------------------|------|-----------|-----------------------|-----------------------|
|        | W/L  | Med. gain | Sign $p$              | Wilcoxon $p$          | W/L  | Med. gain | Sign $p$              | Wilcoxon $p$          |
| 0.1    | 14/1 | 0.0027    | $9.77 \times 10^{-4}$ | $1.22 \times 10^{-4}$ | 14/1 | 2.9742    | $9.77 \times 10^{-4}$ | $8.54 \times 10^{-4}$ |
| 0.3    | 14/1 | 0.0027    | $9.77 \times 10^{-4}$ | $1.22 \times 10^{-4}$ | 13/2 | 1.9194    | 0.007                 | 0.004                 |
| 0.5    | 14/1 | 0.0026    | $9.77 \times 10^{-4}$ | $1.22 \times 10^{-4}$ | 13/2 | 1.6566    | 0.007                 | 0.012                 |
| 0.7    | 14/1 | 0.0027    | $9.77 \times 10^{-4}$ | $1.22 \times 10^{-4}$ | 12/3 | 1.3394    | 0.035                 | 0.048                 |

### 582 C.3 Link Refresh Residual Diagnostic

583 The diagnostic used in the main text reuses the Link Refresh step from Section 5 as a one-round  
584 residual probe. Let  $\hat{\mathcal{Y}}_T$  denote the Tucker output and  $\mathcal{Y}$  the target tensor. Starting from this fitted  
585 Tucker latent tensor, we update only the scalar polynomial coefficients by fitting

$$h_\gamma(x) = \sum_{p=0}^4 \gamma_p x^p$$

586 by ridge regression on pairs  $(\hat{Y}_{T,i}, Y_i)$ . In full reconstruction, all entries are used for this fit. In  
587 completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted  
588 map is applied entrywise to  $\hat{\mathcal{Y}}_T$ . We summarize this cheap refresh by the Link Refresh diagnostic

$$D_{\text{link}} = 10 \log_{10} \left( \frac{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_T\|_F^2}{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{link}}\|_F^2} \right),$$

589 which measures, on a log scale, how much Tucker residual energy is removed on the task-specific  
590 evaluation entries. In other words, it measures how much shared scalar-link yield is already present  
591 before the full NTD-PL alternating solver starts reshaping the Tucker parameters.

### 592 C.4 CAVE Completion Scene Diagnostics

593 Figure 7 supplements the main CAVE diagnostic table with the scene-level scatter plots used to  
594 visualize the link yield across missing rates. The plots show the same diagnostic relationship between  
595  $D_{\text{link}}$  and held-out RMSE\* gain, now at the scene level.

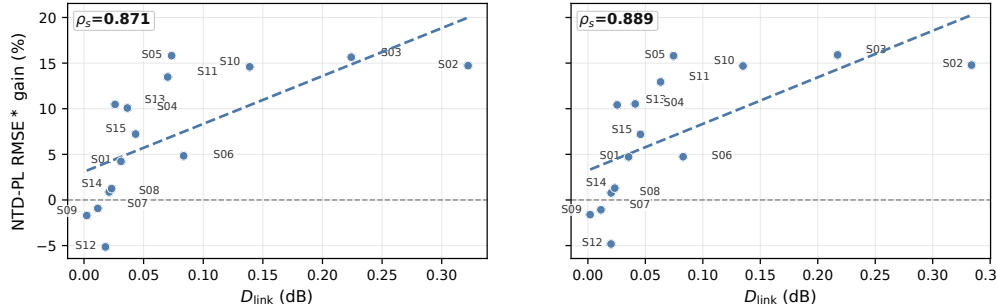


Figure 7: Scene-level supplement for the CAVE completion diagnostic table. The left and right panels use missing rates 0.3 and 0.5, respectively;  $D_{\text{link}}$  from frozen-backbone Link Refresh predicts larger held-out RMSE\* gains.

### 596 C.5 Non-HSI Link Refresh Diagnostics

597 The main diagnostic figure uses per-unit non-HSI experiments rather than collection-level tensors.  
598 CBSD68 [Martin et al., 2001] uses per-image units resized to  $96 \times 96 \times 3$  with rank (32, 32, 3).  
599 CIFAR-10 [Krizhevsky, 2009] uses 1000 class-balanced test images of shape  $32 \times 32 \times 3$  with rank  
600 (20, 20, 3). COIL-100 [Nene et al., 1996] uses one object at a time as a  $36 \times 48 \times 48 \times 3$  view tensor  
601 with rank (8, 12, 12, 2). smallNORB [LeCun et al., 2004] uses one object instance at a time as an

Table 7: Dataset-level statistics behind the non-HSI diagnostic scatter in Figure 4. The gain columns report mean NTD-PL RMSE improvement over independently fitted matched-rank Tucker within  $D_{\text{link}}$  terciles.

| Domain         | Dataset    | Units | Med. $D_{\text{link}}$ | $\rho_s$ | Low    | Mid   | High   |
|----------------|------------|-------|------------------------|----------|--------|-------|--------|
| Natural images | CBSD68     | 68    | 0.091                  | 0.85     | 4.17%  | 8.40% | 14.60% |
| Natural images | CIFAR-10   | 1000  | 0.008                  | 0.93     | -0.01% | 2.59% | 7.85%  |
| Object-view    | COIL-100   | 100   | 0.208                  | 0.93     | 2.77%  | 5.46% | 9.14%  |
| Object-view    | smallNORB  | 25    | 0.264                  | 0.95     | 6.92%  | 9.91% | 14.16% |
| Action video   | KTH-Action | 240   | 0.107                  | 0.88     | 2.74%  | 4.27% | 10.87% |
| Action video   | UCF101     | 240   | 0.065                  | 0.89     | 1.44%  | 2.97% | 5.89%  |

602  $18 \times 64 \times 64 \times 2$  stereo view tensor with rank (12, 24, 24, 2). KTH-Action [Schuldt et al., 2004]  
603 and UCF101 [Soomro et al., 2012] provide the action-video tensor units used in the main diagnostic  
604 scatter. All units are max-normalized independently, and diagnostic labels are assigned only from  
605  $D_{\text{link}}$ , not from NTD-PL gain.

606 **Compute resources.** The reported runs used a CPU Python implementation based on NumPy,  
607 SciPy, and TensorLy on a workstation with a 13th Gen Intel Core i5-13400F CPU (10 physical cores,  
608 16 logical workers), 32 GB system RAM, Windows 11, and Python 3.13. An NVIDIA RTX 5060 GPU  
609 with 8 GB memory was present, but the reported scripts do not use GPU kernels and do not require  
610 GPU memory. Sweep scripts cap BLAS/OpenMP threads per worker and parallelize independent  
611 jobs, using up to 12 workers for the main low-rank sweeps and 6 workers for the diagnostic batches.  
612 From the recorded runtime fields, controlled synthetic residual jobs were sub-second to about 1 s per  
613 configuration. Full-resolution CAVE reconstruction took about 10–17 s per scene for matched-rank  
614 Tucker and 172–217 s per scene for NTD-PL in the main low-rank reconstruction sweep. CAVE  
615 random completion runs took about 0.9–18.6 s per scene/mask/seed (3.3–18.6 s for NTD-PL), while  
616 structured completion runs were about 1.6–5.0 s. The frozen-backbone CAVE Link Refresh diagnostic  
617 itself took about 0.12–0.59 s per fitted refresh; the associated full NTD-PL completion fits used for  
618 comparison were on the order of 168–258 s in the recorded diagnostic runs. Non-HSI diagnostic  
619 batches were run with 6 CPU workers: the CBSD68/CIFAR-10/COIL-100/smallNORB batch took  
620 about 77 s wall-clock, and the KTH-Action and UCF101 video batches took about 5–6 minutes  
621 each. Per-unit NTD-PL runtimes in those diagnostics ranged from about 0.07–0.28 s for image  
622 units, 1.2–2.7 s for object-view units, and 6.3–9.8 s for action-video units. Peak memory was not  
623 instrumented per run; empirically the largest reported CPU runs completed within the 32 GB system  
624 RAM budget.

## 625 C.6 Dataset Asset and Usage Information

626 All datasets used in the experiments are public research benchmarks. We use them only for reconstruc-  
627 tion, completion, or diagnostic evaluation, and the anonymous supplement redistributes processed  
628 summaries and scripts rather than raw dataset files.



Table 8: Dataset sources and usage notes. “No” in the redistribution column means that raw data files are not included in the supplement.

| Dataset                           | Source / citation                                                                          | License or usage note                                                                                                                                                             | Redistribution / use                                               |
|-----------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| CAVE Multispectral Image Database | Columbia CAVE Laboratory repository; [Columbia CAVE Laboratory, n.d., Yasuma et al., 2010] | Official page provides public download and citation information, but does not specify a standard license in the page metadata available to us. Used only for research evaluation. | No; CAVE reconstruction, completion, and Link Refresh diagnostics. |
| CBSD68 / Berkeley natural images  | Berkeley Segmentation Dataset page; [Martin et al., 2001]                                  | Official BSDS usage statement permits use for research and non-commercial purposes only and requests citation of the Berkeley dataset.                                            | No; non-HSI diagnostic evaluation.                                 |
| CIFAR-10                          | University of Toronto CIFAR page; [Krizhevsky, 2009]                                       | Official page provides public research benchmark downloads and citation, but does not specify a standard license in the page metadata available to us.                            | No; non-HSI diagnostic evaluation.                                 |
| COIL-100                          | Columbia Object Image Library page; [Nene et al., 1996]                                    | Official page provides public download and citation information, but does not specify a standard license in the page metadata available to us. Used only for research evaluation. | No; non-HSI diagnostic evaluation.                                 |
| smallNORB                         | NYU/Yann LeCun dataset page; [LeCun et al., 2004]                                          | Official page states that the database is provided for research purposes, cannot be sold, and should be cited when used in publications.                                          | No; non-HSI diagnostic evaluation.                                 |
| KTH-Action                        | KTH action-recognition dataset page; [Schuldt et al., 2004]                                | Official page states that the database is publicly available for non-commercial use and requests citation of the ICPR 2004 paper.                                                 | No; non-HSI diagnostic evaluation.                                 |
| UCF101                            | UCF CRCV dataset page; [Soomro et al., 2012]                                               | Official page provides public benchmark downloads and citation, but does not specify a standard license in the page metadata available to us. Used only for research evaluation.  | No; non-HSI diagnostic evaluation.                                 |

## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes].

Justification: The abstract and introduction state the model, theory, optimizer, and scoped empirical claim. They explicitly frame NTD-PL as a Tucker enhancement for coherent nonlinear residuals, not as a universal tensor decoder.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: Section 7 discusses the scalar-link assumption, rank-budget scope, polynomial-degree stability, nonconvexity, and empirical scope.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes].

Justification: The main theory statements list their assumptions in Section 4; Appendix B gives full proofs and the auxiliary lemma used by the separation result.

### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes].

Justification: Sections 6 and Appendix C.1–C.5 describe datasets, ranks, mask protocols, metrics, diagnostics, and optimization details needed to reproduce the reported comparisons.

### 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Partial].

Justification: The anonymous supplement provides the NTD-PL implementation, baseline wrappers, environment files, command scripts, plotting utilities, and processed CSV summaries for the controlled synthetic, CAVE reconstruction/completion, and Link Refresh diagnostic results. Public datasets are not redistributed; full non-HSI diagnostic reruns require separate dataset downloads and are documented with smoke/full scripts.

## 6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: The experiment section and appendix specify the main ranks, mask rates, held-out metrics, datasets, and optimizer structure; Appendix A.2 gives the update equations and algorithm.

## 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes].

Justification: The CAVE completion table reports mean and standard deviation over scenes, and Appendix C.2 reports paired sign and Wilcoxon tests for the same protocol.

## 8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Partial].

Justification: Appendix C.5 reports the CPU/RAM/GPU environment, CPU-only implementation note, parallel worker settings, and approximate runtime ranges from recorded fit-time and diagnostic wall-clock fields. Peak memory was not instrumented per run, so memory is reported as an observed machine-level bound rather than exact per-experiment usage.

## 9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes].

Justification: The work uses mathematical analysis and public benchmark data for tensor modeling, and no human-subjects study, deceptive deployment, or high-risk data release is introduced.

## 10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A].

Justification: The contribution is a foundational tensor-modeling method with experiments on standard public benchmarks; it does not introduce a deployed system, user-facing model, or application-specific decision pipeline.

## 11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A].

Justification: The paper does not release a high-risk pretrained model, generative model, or newly scraped dataset requiring special safeguards.

## 12. Licenses for existing assets

709 Question: Are the creators or original owners of assets (e.g., code, data, models), used in  
710 the paper, properly credited and are the license and terms of use explicitly mentioned and  
711 properly respected?

712 Answer: **[Partial]**.

713 Justification: Appendix C.6 lists the official source, citation, usage note or license infor-  
714 mation, redistribution status, and experimental use for each dataset. Raw datasets are not  
715 redistributed in the supplement. Some official dataset pages provide research-use or citation  
716 guidance but do not specify a standard license, so we mark this item as partial rather than  
717 full.

718 **13. New assets**

719 Question: Are new assets introduced in the paper well documented and is the documentation  
720 provided alongside the assets?

721 Answer: **[N/A]**.

722 Justification: The paper introduces a method and experiments, not a new dataset, model  
723 checkpoint, or benchmark asset.

724 **14. Crowdsourcing and research with human subjects**

725 Question: For crowdsourcing experiments and research with human subjects, does the paper  
726 include the full text of instructions given to participants and screenshots, if applicable, as  
727 well as details about compensation (if any)?

728 Answer: **[N/A]**.

729 Justification: The paper does not involve crowdsourcing or human-subjects experiments.

730 **15. Institutional review board (IRB) approvals or equivalent for research with human**  
731 **subjects**

732 Question: Does the paper describe potential risks incurred by study participants, whether  
733 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)  
734 approvals (or an equivalent approval/review based on the requirements of your country or  
735 institution) were obtained?

736 Answer: **[N/A]**.

737 Justification: The paper does not involve human-subjects research.

738 **16. Declaration of LLM usage**

739 Question: Does the paper describe the usage of LLMs if it is an important, original, or  
740 non-standard component of the core methods in this research? Note that if the LLM is used  
741 only for writing, editing, or formatting purposes and does *not* impact the core methodology,  
742 scientific rigor, or originality of the research, declaration is not required.

743 Answer: **[N/A]**.

744 Justification: The core method, theory, and experiments do not use LLMs as an important,  
745 original, or non-standard research component.